IntegrationPlan.md

LLM-Mamba4Rec Integration Architecture Plan

This document outlines the integration strategy for combining an LLM-based conversational interface with the Mamba4Rec sequential recommendation system for a movie recommender application.

a. BPR vs CE Loss for LLM Integration

Recommendation: Use Cross-Entropy (CE) Loss

Rationale:

1. **Full Softmax Distribution**: CE loss computes a probability distribution over ALL items, which is essential when LLM mood vectors need to influence rankings globally. The softmax ensures that when a user says "I want something Tarantino-style," the model can appropriately redistribute probability mass across all Tarantino-esque films.
2. **Gradient Flow to All Items**: With CE, gradients flow through the entire item embedding matrix during each update. This means the LLM projection layer learns how its output affects scores for every item, not just a sampled pair. This is critical for mood-based adjustments that should affect genre clusters, directors, eras, etc.

3. **Better Calibration**: CE loss produces better-calibrated probability estimates, which matters when you want to explain recommendations to users ("This movie scored 85% match with your current mood").
4. **Contrastive Limitations of BPR**: BPR only considers (positive, negative) pairs and doesn't naturally support the soft preference shifts that LLM mood vectors introduce. A user saying "cozy but not too long" isn't a hard positive/negative distinction—it's a preference modulation.

Trade-off: CE is computationally more expensive (full softmax over all items), but for movie datasets like ML-1M (~3,900 items), this is tractable.

b. Is Gated Fusion the Best Approach?

Yes, gated fusion is well-suited for this use case. Here's why:

Advantages of Gated Fusion

1. **Adaptive Blending**: The gate learns when to trust the Mamba sequence history vs. the LLM mood signal. If a user has a strong viewing pattern and a vague mood ("something good"), the gate can weight history heavily. If the mood is specific ("1970s horror"), the gate can shift toward the LLM signal.
2. **Graceful Degradation**: When no LLM signal is present (m_current = 0), the gate naturally learns to output $\alpha \approx 1$, preserving pure Mamba behavior. This makes the system robust during Phase 1 training and when users don't specify preferences.
3. **Avoids Hard Switching**: Unlike attention mechanisms that might entirely ignore one signal, gated fusion ensures both signals contribute (to varying degrees).

Alternative Approaches Considered

Approach	Pros	Cons
Concatenation + MLP	Simple, expressive	Doesn't preserve Mamba geometry; requires more training data
Additive Fusion	Very simple, preserves geometry	No learnable weighting; can't handle conflicting signals
Cross-Attention	Powerful for complex interactions	Computationally expensive; overkill for scalar mood vectors
Residual Addition	Minimal interference	No adaptation to signal strength

Recommendation

Keep gated fusion but consider a refinement: **vector-level gating** instead of scalar gating. The current implementation uses a single $\alpha \in [0,1]$ for the entire hidden dimension. A per-dimension gate $\alpha \in \mathbb{R}^d$ would allow the model to trust Mamba for some latent dimensions (e.g., genre preferences) while trusting LLM for others (e.g., mood modifiers).

c. Is the Current Code Setup Sufficient for Fusion?

No, several components are missing or need modification.

Current State Analysis

What's Working:

- `PreferenceFusion` module in `fusion.py` implements gated fusion correctly
- `mamba4rec.py` has conditional fusion in `predict()` and `full_sort_predict()`
- Config flag `use_llm_fusion` allows toggling fusion on/off

What's Missing:

1. **Import Statement**: `mamba4rec.py` uses `PreferenceFusion` but doesn't import it:

```
# Missing at top of mamba4rec.py:
from fusion import PreferenceFusion
```

2. **LLMProjection Integration**: `LLMprojection.py` defines the projection module but:
   - Missing `import torch.nn as nn`
   - Not integrated into the main model
   - Not instantiated anywhere
3. **Data Pipeline for LLM Embeddings**: The `interaction` dictionary expects `LLM_MOOD_EMB` and `LLM_PROFILE_EMB` keys, but:
   - No data loader modifications to include these
   - No mechanism to generate LLM embeddings during training/inference

- No storage schema for user mood vectors

4. **Training Loss Doesn't Flow Through Fusion**: `calculate_loss()` doesn't use fusion—only `predict()` and `full_sort_predict()` do. This means fusion isn't trained during the main training loop.
5. **Freezing Logic Missing**: The phased training strategy requires freezing specific layers, but no freeze/unfreeze utilities exist.

Required Additions

```
Required New Files/Modifications:
├── llm_encoder.py          # Wrapper for sentence transformer
├── embedding_store.py      # Vector database interface
├── dataset_wrapper.py      # Modified data loader
├── train_phases.py         # Phased training orchestrator
└── inference_server.py     # Real-time fusion inference
```

d. Is the Proposed Training Plan Optimal?

The plan is sound but needs refinements for your use case.

Original Plan Assessment

Phase	Assessment	Modification
Phase 1: Train vanilla Mamba4Rec	✓ Correct	None needed
Phase 2: Freeze Mamba, train LLM projection + fusion	✓ Correct concept	Add warm-up scheduling

Phase	Assessment	Modification
Phase 3: Joint fine-tuning	⚠ Optional but risky	Add constraints

Recommended Modifications

Phase 2 Refinements

1. **Contrastive Pre-alignment**: Before training LLMProjection with recommendation loss, pre-align LLM embeddings to item embeddings using contrastive learning:

   ```
   For movie "Pulp Fiction":
   - Positive pairs: ("Pulp Fiction" sentence embedding, item embeddin
   - Negative pairs: (random movie description, item embedding)
   ```

   This gives the projection a better initialization.
2. **Curriculum Learning**: Start with "easy" mood signals (explicit genre requests like "action movie") before introducing nuanced moods ("something like early Coen brothers but lighter").
3. **Gradient Clipping**: Apply aggressive gradient clipping to prevent the LLM projection from dominating early training.

Phase 3 Constraints

1. **Elastic Weight Consolidation (EWC)**: When unfreezing, use EWC to penalize large changes to parameters that were important for Phase 1 performance.
2. **Very Low Learning Rate**: Use 1/10 to 1/100 of Phase 2 learning rate.
3. **Early Stopping on Sequence Metrics**: Monitor Hit@10 on sequential evaluation (without LLM signals). If it drops more than 5%, stop fine-tuning.

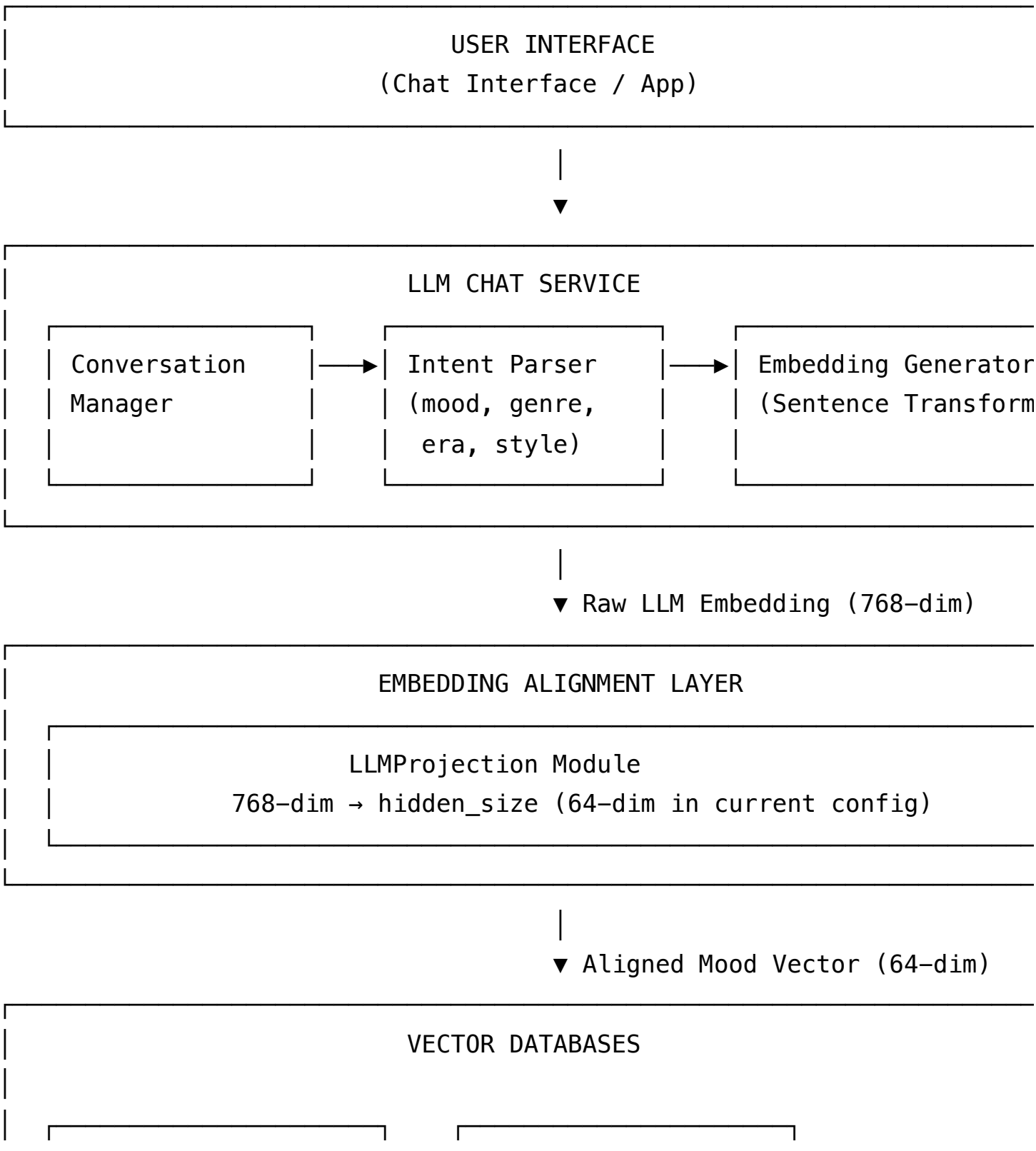
Additional Phase: Profile Learning

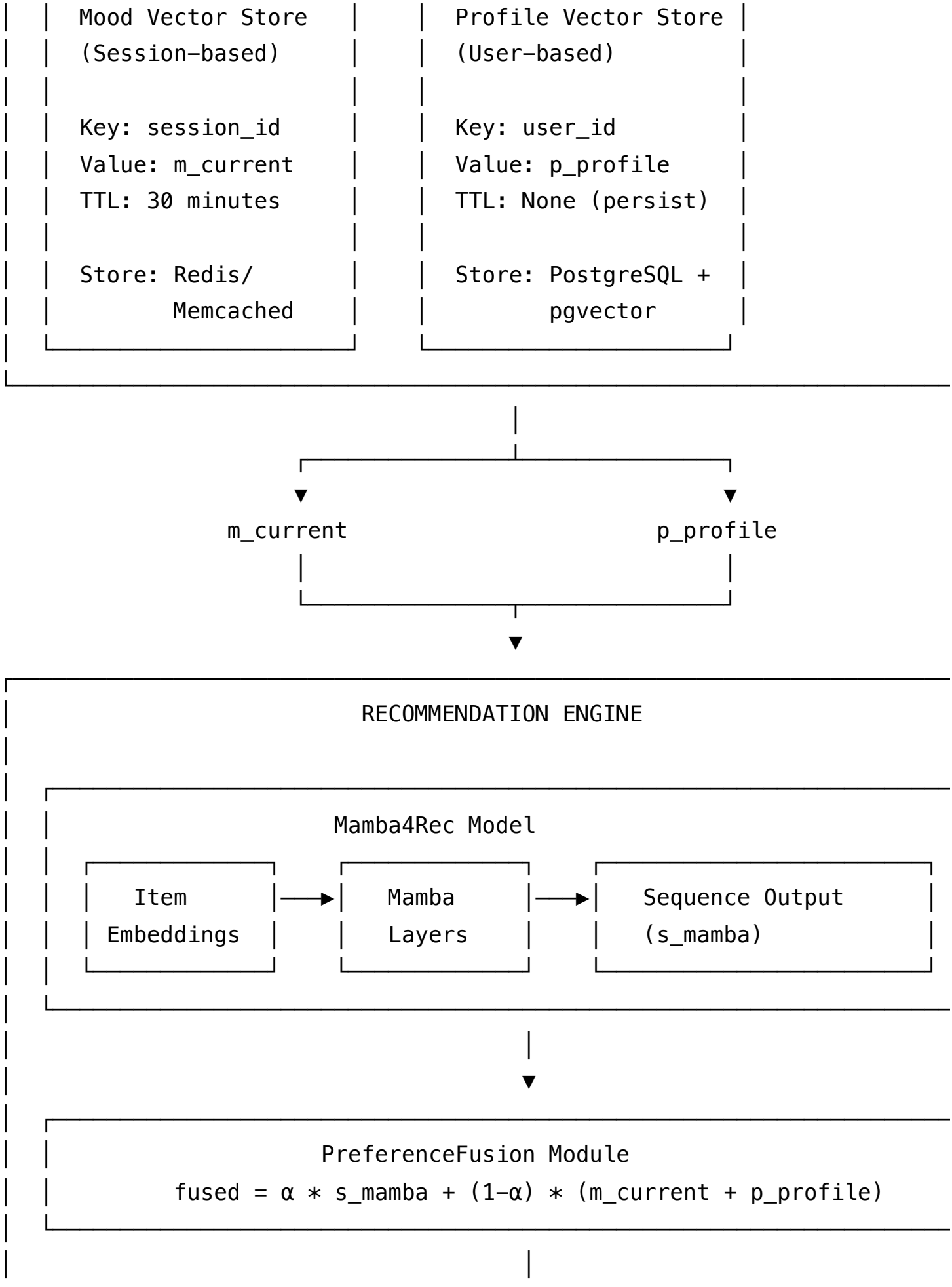
Consider adding **Phase 2.5**:

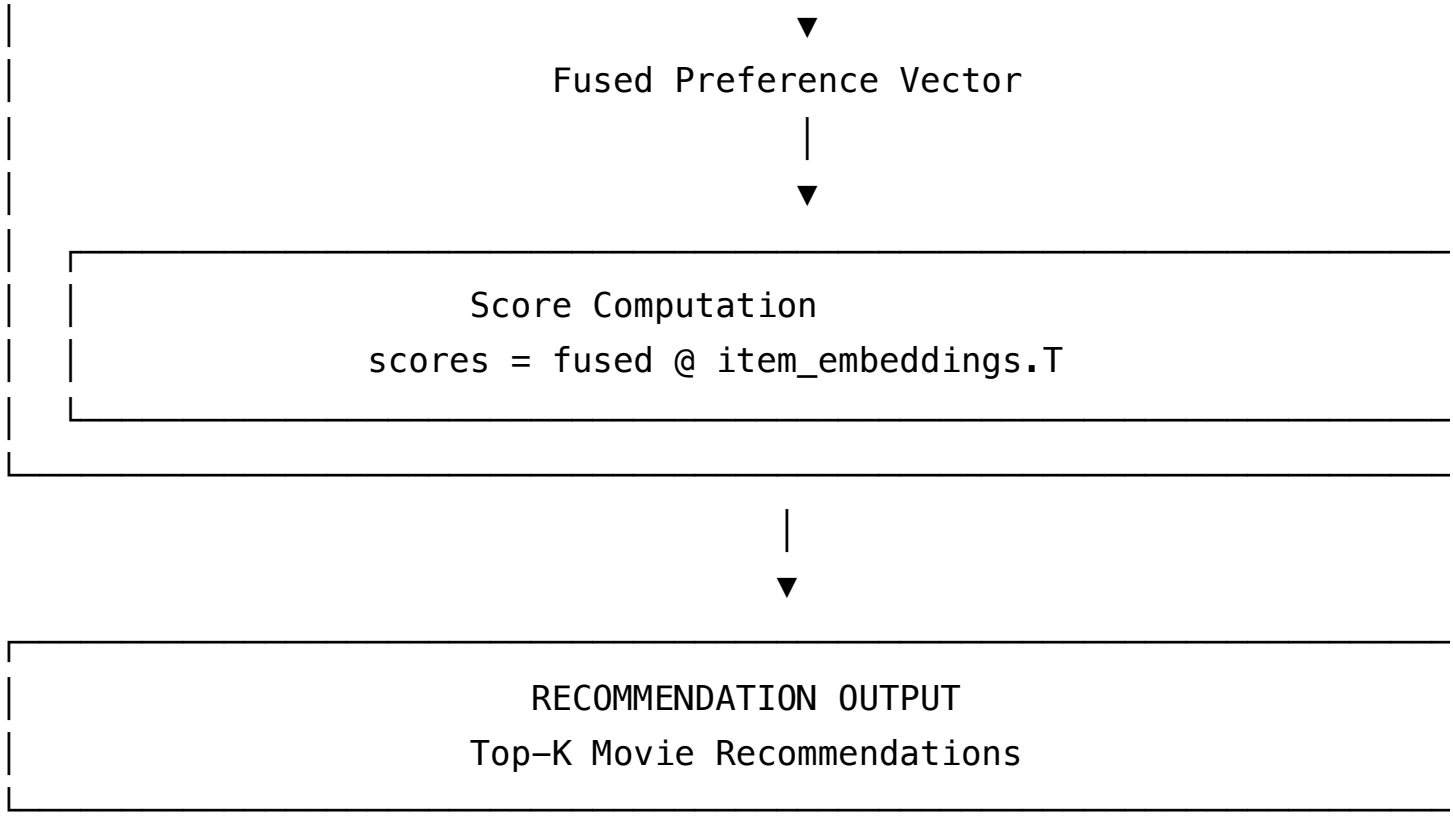
- Freeze: item embeddings, Mamba layers, LLMProjection
- Train: Profile aggregation layer (p_profile)
- Purpose: Learn stable user profiles from historical LLM interactions

e. Overall Architecture for LLM Integration

System Architecture Diagram




Mood Vector Store
(Session-based)
Key: session_id
Value: m_current
TTL: 30 minutes
Store: Redis/
Memcached
Profile Vector Store
(User-based)
Key: user_id
Value: p_profile
TTL: None (persist)
Store: PostgreSQL +
pgvector
m_current
p_profile
RECOMMENDATION ENGINE
Mamba4Rec Model
Item
Embeddings
Mamba
Layers
Sequence Output
(s_mamba)
PreferenceFusion Module
fused = α * s_mamba + (1−α) * (m_current + p_profile)



Database Schema

1. Mood Vector Store (Redis)

```
Key Structure: mood:{session_id}
Value: JSON {
    "vector": [float; 64],
    "raw_text": "I want something cozy from the early 2000s",
    "parsed_intent": {
        "mood": "cozy",
        "era": "2000-2010",
        "genre": null
    },
    "timestamp": "2024-01-15T10:30:00Z"
}
TTL: 1800 seconds (30 minutes)
```

2. Profile Vector Store (PostgreSQL + pgvector)

```
CREATE TABLE user_profiles (
    user_id         BIGINT PRIMARY KEY,
    profile_vector  VECTOR(64),          -- p_profile
    interaction_count INT DEFAULT 0,
    last_updated    TIMESTAMP,
    created_at      TIMESTAMP DEFAULT NOW()
);

CREATE TABLE mood_history (
    id              SERIAL PRIMARY KEY,
    user_id         BIGINT REFERENCES user_profiles(user_id),
    session_id      UUID,
    mood_vector     VECTOR(64),
    raw_text        TEXT,
    feedback_signal FLOAT,               -- Did user like the recommend
    created_at      TIMESTAMP DEFAULT NOW()
);

CREATE INDEX idx_mood_history_user ON mood_history(user_id, created_at
```

3. Item Embeddings Cache (Redis)

```
Key Structure: item_emb:{item_id}
Value: [float; 64]
TTL: None (permanent, refresh on model update)
```

LLM Vector Alignment Strategy

The key challenge is ensuring LLM embeddings (typically 768 or 1024 dimensions from sentence transformers) align with Mamba's learned item geometry (64

dimensions).

Alignment Approach: Anchored Projection

1. **Anchor Points**: Use movie descriptions to create alignment anchors:

```
# For each movie in the catalog:
movie_description = "Pulp Fiction (1994): Interconnected crime stor
llm_embedding = sentence_transformer.encode(movie_description)  # 7
item_embedding = mamba_model.item_embedding(item_id)  # 64-dim

# These pairs become training data for LLMProjection
```

2. **Contrastive Alignment Loss**:

```
def alignment_loss(projected_llm, item_emb, temperature=0.07):
    # Cosine similarity
    sim = F.cosine_similarity(projected_llm, item_emb)

    # Contrastive loss with in-batch negatives
    logits = sim / temperature
    labels = torch.arange(len(sim))
    return F.cross_entropy(logits, labels)
```

3. **Geometric Preservation**: During projection training, add a regularization term that preserves relative distances:

```
def geometry_preservation_loss(projected_1, projected_2, original_1
    proj_dist = F.pairwise_distance(projected_1, projected_2)
    orig_dist = F.pairwise_distance(original_1, original_2)
    return F.mse_loss(proj_dist, orig_dist)
```

Updating Vectors with New Interactions

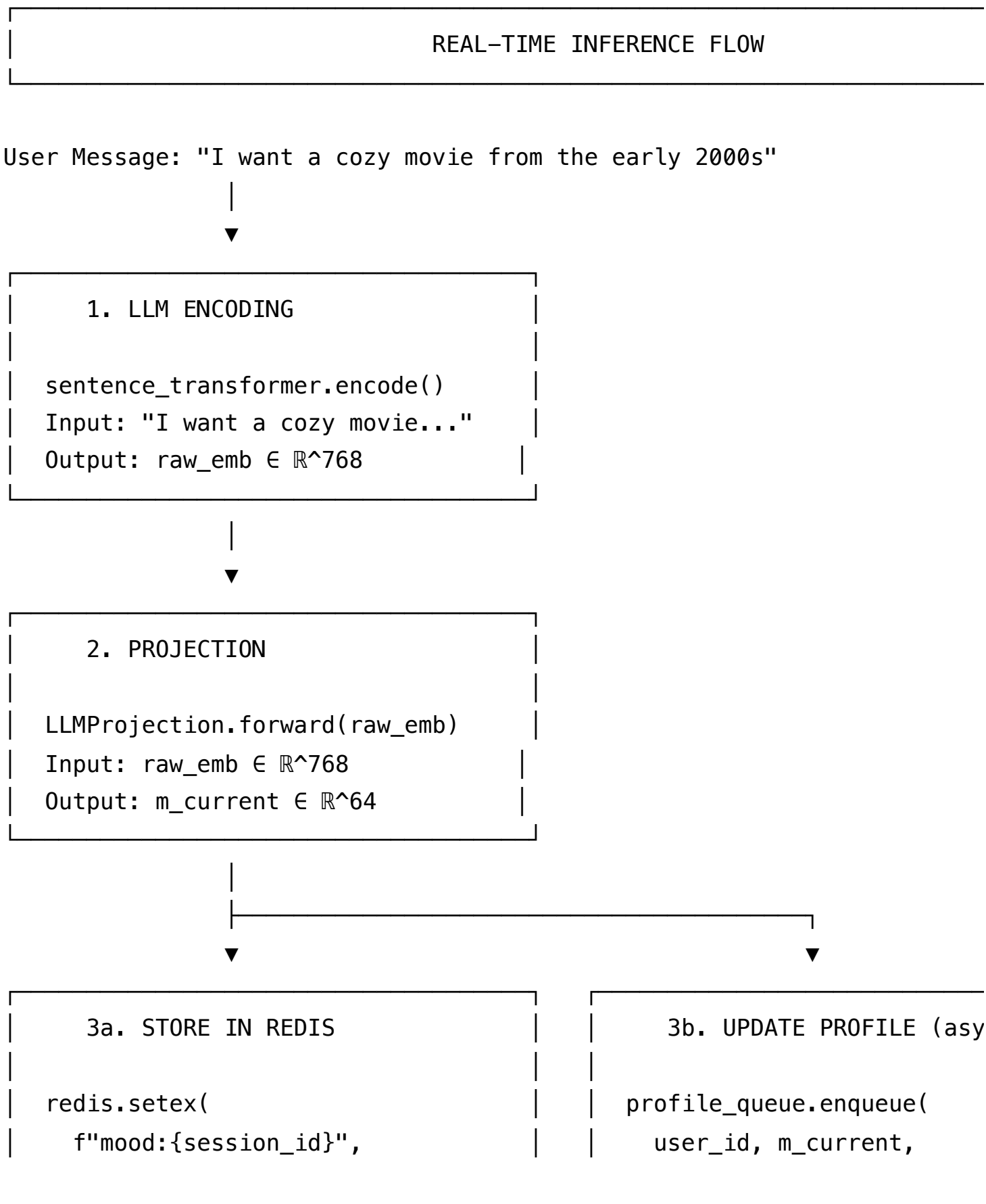
Real-time Mood Updates

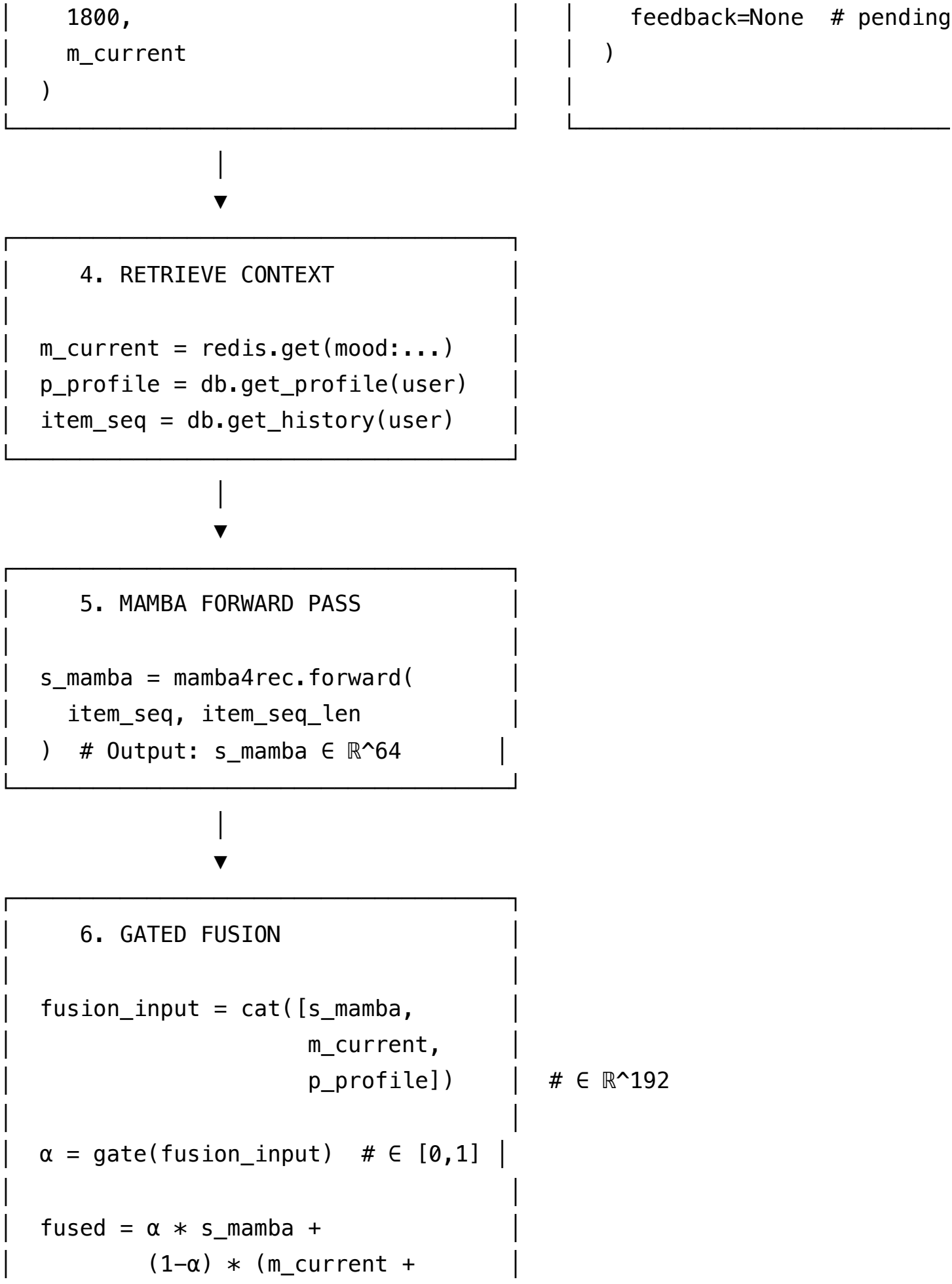
```
def update_mood_vector(session_id, user_message, llm_encoder, projectio
    # 1. Encode the new message
    raw_embedding = llm_encoder.encode(user_message)

    # 2. Project to Mamba space
    aligned_embedding = projection(raw_embedding)

    # 3. Retrieve existing mood (if any)
    existing_mood = redis.get(f"mood:{session_id}")

    if existing_mood:
        # Exponential moving average for conversation continuity
        alpha = 0.7  # Weight toward new message
        new_mood = alpha * aligned_embedding + (1 - alpha) * existing_m
    else:
        new_mood = aligned_embedding

    # 4. Store updated mood
    redis.setex(f"mood:{session_id}", 1800, new_mood)

    return new_mood
```

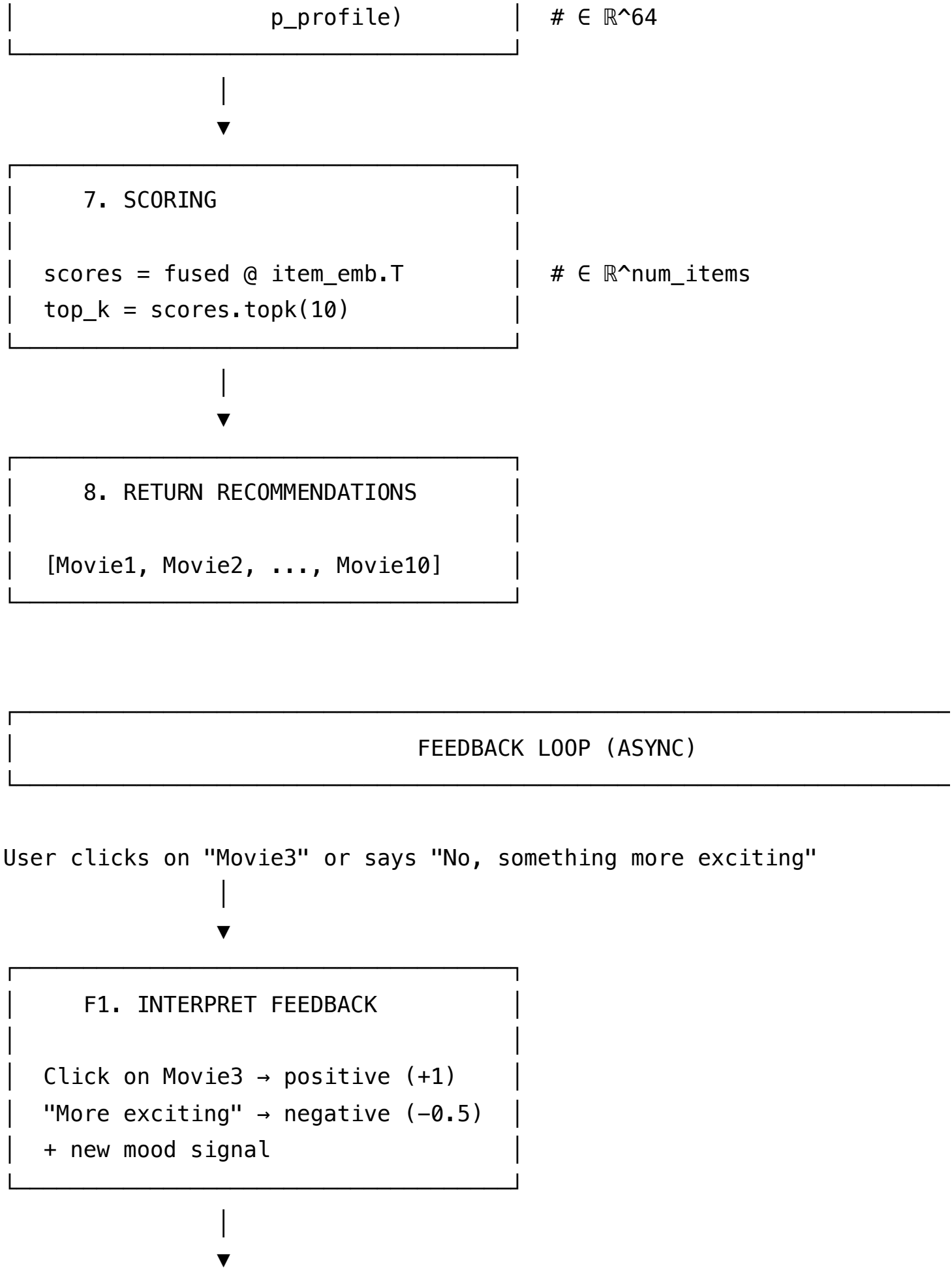
Long-term Profile Updates

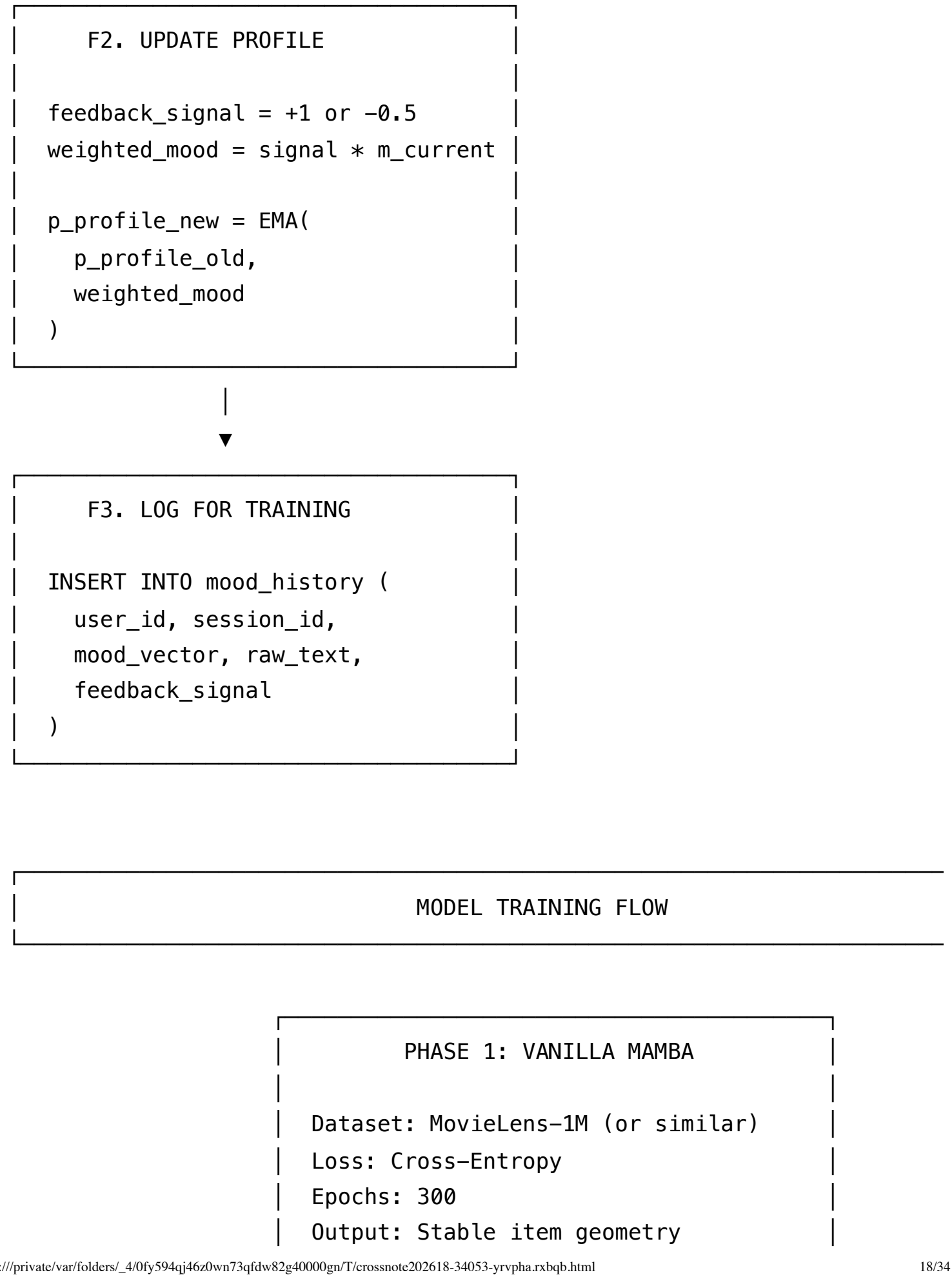
```
def update_user_profile(user_id, mood_vector, feedback_signal, db):
    """
    feedback_signal: 1.0 if user liked recommendation, -1.0 if disliked
    """
    # 1. Retrieve current profile
    current_profile = db.get_profile(user_id)

    # 2. Weight the mood by feedback
    weighted_mood = feedback_signal * mood_vector

    # 3. Update with exponential moving average
    decay = 0.95  # Slow decay for long-term preferences
    interaction_count = db.get_interaction_count(user_id)

    # Adaptive learning rate: learn faster early, slower later
    lr = 0.1 / (1 + 0.01 * interaction_count)

    new_profile = decay * current_profile + lr * weighted_mood

    # 4. Normalize to unit sphere (prevents drift)
    new_profile = F.normalize(new_profile, dim=-1)

    # 5. Store
    db.update_profile(user_id, new_profile)
    db.log_mood_history(user_id, mood_vector, feedback_signal)
```

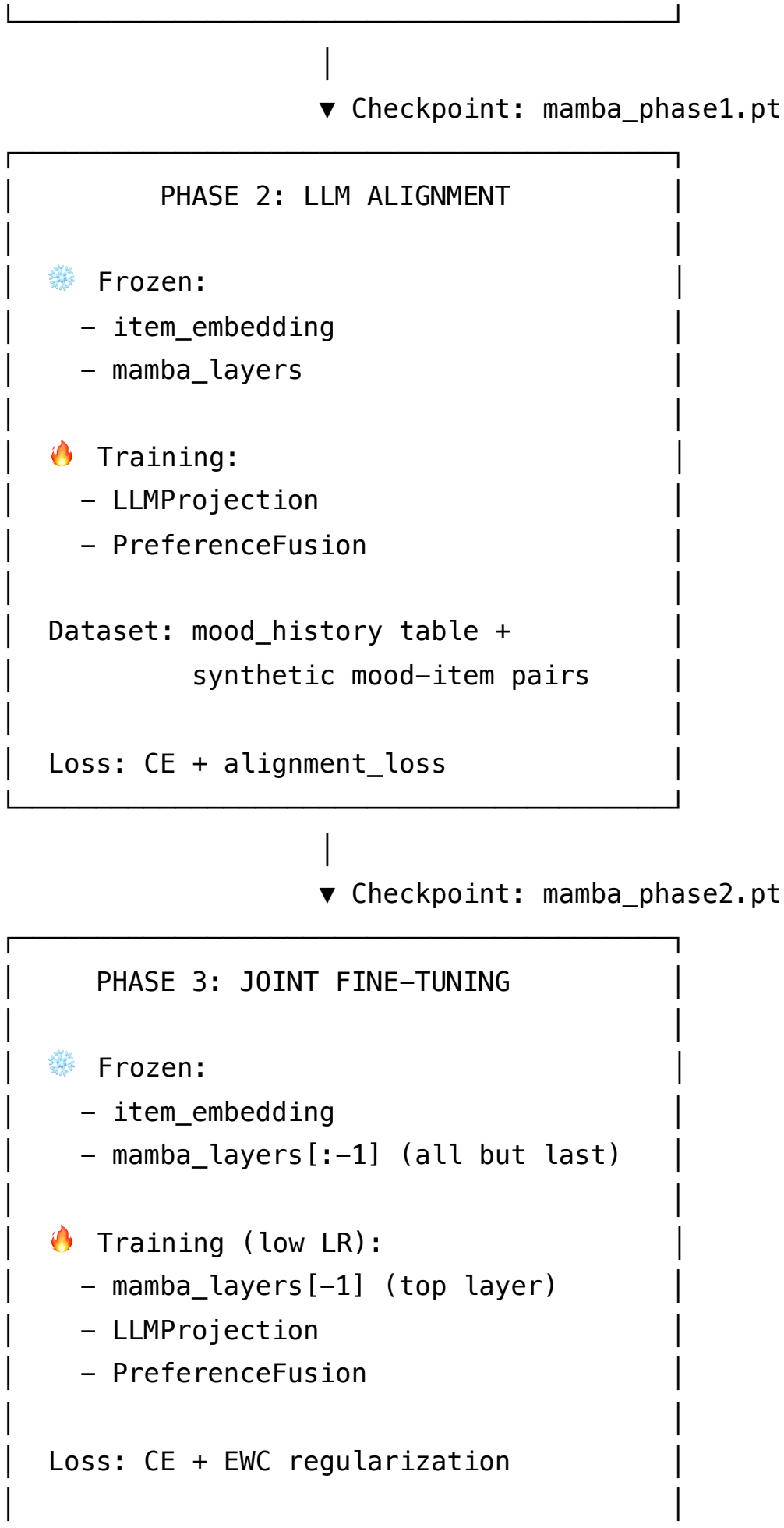
f. Data Flow Diagram




1800,
m_current
)
feedback=None # pending
)
4. RETRIEVE CONTEXT
m_current = redis.get(mood:...)
p_profile = db.get_profile(user)
item_seq = db.get_history(user)
5. MAMBA FORWARD PASS
s_mamba = mamba4rec.forward(
item_seq, item_seq_len
) # Output: s_mamba ∈ ℝ^64
6. GATED FUSION
fusion_input = cat([s_mamba,
m_current,
p_profile]) # ∈ ℝ^192
α = gate(fusion_input) # ∈ [0,1]
fused = α * s_mamba +
(1-α) * (m_current +


p_profile)
∈ ℝ^64
7. SCORING
scores = fused @ item_emb.T
# ∈ ℝ^num_items
top_k = scores.topk(10)
8. RETURN RECOMMENDATIONS
[Movie1, Movie2, ..., Movie10]
FEEDBACK LOOP (ASYNC)
User clicks on "Movie3" or says "No, something more exciting"
F1. INTERPRET FEEDBACK
Click on Movie3 → positive (+1)
"More exciting" → negative (−0.5)
+ new mood signal


F2. UPDATE PROFILE
feedback_signal = +1 or -0.5
weighted_mood = signal * m_current
p_profile_new = EMA(
p_profile_old,
weighted_mood
)
F3. LOG FOR TRAINING
INSERT INTO mood_history (
user_id, session_id,
mood_vector, raw_text,
feedback_signal
)
MODEL TRAINING FLOW
PHASE 1: VANILLA MAMBA
Dataset: MovieLens-1M (or similar)
Loss: Cross-Entropy
Epochs: 300
Output: Stable item geometry


▼ Checkpoint: mamba_phase1.pt
PHASE 2: LLM ALIGNMENT
❄️ Frozen:
- item_embedding
- mamba_layers
🔥 Training:
- LLMProjection
- PreferenceFusion
Dataset: mood_history table +
synthetic mood-item pairs
Loss: CE + alignment_loss
▼ Checkpoint: mamba_phase2.pt
PHASE 3: JOINT FINE-TUNING
❄️ Frozen:
- item_embedding
- mamba_layers[:-1] (all but last)
🔥 Training (low LR):
- mamba_layers[-1] (top layer)
- LLMProjection
- PreferenceFusion
Loss: CE + EWC regularization

```
│  Early Stop: if Hit@10 drops > 5%       │
└─────────────────────────────────────────┘
                    │
                    ▼ Checkpoint: mamba_production.

┌─────────────────────────────────────────┐
│        CONTINUOUS LEARNING              │
│             (Optional)                  │
│                                         │
│  Trigger: Every 10K new interactions    │
│                                         │
│  1. Export mood_history since last      │
│     training                            │
│  2. Fine-tune LLMProjection only        │
│  3. A/B test new model                  │
│  4. Promote if metrics improve          │
└─────────────────────────────────────────┘
```

g. Code Changes Documentation

The `Fusion/` folder contains a complete implementation with all necessary modifications for LLM-Mamba integration. Below is a detailed breakdown of each file and the changes made.

File Structure

```
Fusion/
├── config.yaml           # Extended config with fusion settings
├── mamba4rec.py          # Core model with fusion integration
├── fusion.py             # Enhanced fusion module with multiple variant
├── llm_projection.py     # Sophisticated projection with alignment
├── llm_encoder.py        # Sentence transformer wrapper
├── embedding_store.py    # Mood and profile vector storage
├── train_phases.py       # Phased training orchestrator
├── inference.py          # High-level inference API
├── run.py                # Simple training script
├── environment.yaml      # Updated dependencies
└── README.md             # Usage documentation
```

1. `mamba4rec.py` - Core Model Changes

Location: `Fusion/mamba4rec.py`

Key Changes from Original:

a. Added Imports and LLM Components (Lines 1-25)

```
from fusion import PreferenceFusion, AdaptivePreferenceFusion
from llm_projection import LLMProjection, LLMProjectionWithAlignment
```

The original code referenced `PreferenceFusion` but never imported it. This is now fixed.

b. Model Initialization with LLM Components (Lines 50-75)

```
# LLM Fusion components (only created if enabled)
if self.use_llm_fusion:
    self.llm_projection = LLMProjectionWithAlignment(
        llm_dim=self.llm_dim,
        hidden_size=self.hidden_size,
        dropout=self.fusion_dropout
    )
    self.fusion = PreferenceFusion(
        hidden_size=self.hidden_size,
        dropout=self.fusion_dropout,
        vector_gate=self.vector_gate
    )
```

Purpose: Instantiates the LLM projection layer within the model, so raw LLM embeddings (768-dim) are projected to hidden_size (64-dim) before fusion.

c. `_apply_fusion()` Helper Method (Lines 100-125)

```
def _apply_fusion(self, seq_output, interaction):
    if not self.use_llm_fusion or self.fusion is None:
        return seq_output

    raw_mood_emb = interaction.get("LLM_MOOD_EMB", None)
    raw_profile_emb = interaction.get("LLM_PROFILE_EMB", None)

    m_current = None
    p_profile = None

    if raw_mood_emb is not None:
        m_current = self.llm_projection(raw_mood_emb)
    if raw_profile_emb is not None:
        p_profile = self.llm_projection(raw_profile_emb)

    return self.fusion(s_mamba=seq_output, m_current=m_current, p_profi
```

Purpose: Centralizes fusion logic. The original code had fusion calls duplicated in `predict()` and `full_sort_predict()` but not in `calculate_loss()`. This helper ensures consistent application.

d. `calculate_loss()` Now Supports Fusion Training (Lines 130-160)

```
def calculate_loss(self, interaction):
    seq_output = self.forward(item_seq, item_seq_len)

    # Apply fusion during training (Phase 2+)
    if self._current_phase >= 2:
        seq_output = self._apply_fusion(seq_output, interaction)
    ...
```

Purpose: The original code only applied fusion during inference (`predict` , `full_sort_predict`), meaning fusion was never trained. Now fusion is applied during loss calculation in Phase 2+.

e. Phased Training Methods (Lines 190-260)

```
def set_training_phase(self, phase):
    self._current_phase = phase
    if phase == 1:
        self._unfreeze_all()
    elif phase == 2:
        self._freeze_for_phase2()
    elif phase == 3:
        self._configure_phase3()

def _freeze_for_phase2(self):
    # Freeze item embeddings
    self.item_embedding.weight.requires_grad = False
    # Freeze Mamba layers
    for layer in self.mamba_layers:
        for param in layer.parameters():
            param.requires_grad = False
    # Keep fusion trainable
    for param in self.fusion.parameters():
        param.requires_grad = True
```

Purpose: Implements the freezing logic for phased training. The original code had no mechanism to freeze/unfreeze specific layers.

f. Interpretability Method (Lines 265-285)

```
def get_fusion_weights(self, interaction):
    """Get fusion gate values for interpretability."""
    return self.fusion.get_gate_value(seq_output, m_current, p_profile)
```

Purpose: Allows inspection of how much the model trusts Mamba vs. LLM signals for any given prediction.

2. `fusion.py` - Enhanced Fusion Module

Location: `Fusion/fusion.py`

Key Changes from Original:

a. Vector-Level Gating Option (Lines 20-35)

```
def __init__(self, hidden_size, dropout=0.1, vector_gate=False):
    gate_out_dim = hidden_size if vector_gate else 1
    self.gate = nn.Sequential(
        nn.Linear(hidden_size * 3, hidden_size),
        nn.LayerNorm(hidden_size),
        nn.GELU(),
        nn.Dropout(dropout),
        nn.Linear(hidden_size, gate_out_dim),
        nn.Sigmoid()
    )
```

Purpose: Original fusion used scalar gating (α ∈ [0,1]). Vector gating (α ∈ ℝ^d) allows the model to trust different signals for different latent dimensions.

b. Gate Bias Initialization (Lines 45-55)

```
def _init_gate_bias(self):
    """Initialize gate to slightly favor Mamba output (α ≈ 0.6)."""
    with torch.no_grad():
        final_linear = self.gate[-2]
        if hasattr(final_linear, 'bias') and final_linear.bias is not N
            final_linear.bias.fill_(0.4)
```

Purpose: Biases the gate toward Mamba at initialization, ensuring stable behavior when LLM signals are weak or absent.

c. Additional Fusion Variants (Lines 100-200)

- `AdaptivePreferenceFusion` : Attention-based fusion with multi-head attention
- `TemporalPreferenceFusion` : Processes mood history with a transformer for conversation context

Purpose: Provides alternatives for experimentation and more complex use cases.

3. `llm_projection.py` - Sophisticated Projection Module

Location: `Fusion/llm_projection.py`

Key Changes from Original `LLMprojection.py` :

a. Fixed Missing Import

```
import torch.nn as nn  # Was missing in original
```

b. Multi-Layer Architecture with Normalization (Lines 30-55)

```
intermediate_dim = max(hidden_size * 2, llm_dim // 4)

self.proj = nn.Sequential(
    nn.Linear(llm_dim, intermediate_dim),
    nn.LayerNorm(intermediate_dim),
    nn.GELU(),
    nn.Dropout(dropout),
    nn.Linear(intermediate_dim, hidden_size),
    nn.LayerNorm(hidden_size),
    nn.GELU(),
    nn.Dropout(dropout),
    nn.Linear(hidden_size, hidden_size),
)
```

Purpose: Original was 2 layers (768→64→64). New version uses a bottleneck design with layer normalization for training stability.

c. Small Weight Initialization (Lines 60-70)

```
def _init_weights(self, module):
    if isinstance(module, nn.Linear):
        nn.init.normal_(module.weight, mean=0.0, std=0.01)  # Smaller t
```

Purpose: Prevents the LLM projection from dominating Mamba outputs during early training.

d. Contrastive Alignment Methods (Lines 100-160)

```
def compute_alignment_loss(self, llm_emb, item_emb):
    """Contrastive loss to align LLM projections with item embeddings."
    _, contrastive_proj = self.forward(llm_emb, return_contrastive=True

    contrastive_proj = F.normalize(contrastive_proj, dim=-1)
    item_emb = F.normalize(item_emb, dim=-1)

    logits = torch.matmul(contrastive_proj, item_emb.T) / self.temperat
    labels = torch.arange(batch_size, device=llm_emb.device)

    loss_llm_to_item = F.cross_entropy(logits, labels)
    loss_item_to_llm = F.cross_entropy(logits.T, labels)
    return (loss_llm_to_item + loss_item_to_llm) / 2
```

Purpose: Enables pre-alignment of LLM embeddings with item embeddings using movie descriptions as anchor points.

4. `llm_encoder.py` - Sentence Transformer Wrapper

Location: `Fusion/llm_encoder.py`

New File - Not in Original

Key Features:

- **Lazy Loading**: Sentence transformer model is only loaded when first used
- **Caching**: LRU cache for repeated queries (10,000 entries)
- **Mood Preprocessing**: Adds context to short mood messages

- **Intent Parser**: Extracts structured information (mood, genre, era, constraints)

```
class IntentParser:
    MOOD_KEYWORDS = {
        "cozy": ["cozy", "warm", "comfortable", ...],
        "exciting": ["exciting", "thrilling", "action", ...],
        ...
    }

    def parse(self, message):
        return {
            "mood": self._extract_matches(message, self.MOOD_KEYWORDS),
            "genre": self._extract_matches(message, self.GENRE_KEYWORDS
            "era": self._extract_matches(message, self.ERA_KEYWORDS),
            "constraints": self._extract_constraints(message)
        }
```

5. `embedding_store.py` - Vector Storage

Location: `Fusion/embedding_store.py`

New File - Not in Original

Implements:

- `InMemoryMoodStore` : For development/testing with TTL expiration
- `InMemoryProfileStore` : With EMA profile updates
- `RedisMoodStore` : Production-ready Redis backend
- `EmbeddingManager` : Unified interface for all storage operations

```
class EmbeddingManager:
    def update_mood(self, session_id, projected_mood, raw_text, parsed_
        """Update session mood with EMA blending."""
        ...

    def record_feedback(self, user_id, session_id, feedback):
        """Update long-term profile based on feedback."""
        ...

    def prepare_interaction_dict(self, session_id, user_id, device):
        """Create LLM_MOOD_EMB and LLM_PROFILE_EMB for model input."""
        ...
```

6. `train_phases.py` - Phased Training Orchestrator

Location: `Fusion/train_phases.py`

New File - Not in Original

Implements:

- Phase 1: Vanilla Mamba training
- Phase 2: LLM alignment with frozen Mamba
- Phase 3: Joint fine-tuning with EWC regularization

```
class EWCRegularizer:
    """Elastic Weight Consolidation to prevent catastrophic forgetting.

    def _compute_fisher(self, dataloader):
        """Compute Fisher information matrix diagonal."""
        ...

    def penalty(self):
        """Compute EWC penalty term."""
        return self.lambda_ewc * sum(
            (self.fisher[name] * (param - self.optimal_params[name])**2
            for name, param in model.named_parameters()
        )
```

7. `inference.py` - High-Level Inference API

Location: `Fusion/inference.py`

New File - Not in Original

Provides:

- `RecommendationEngine` : Complete inference pipeline
- `RecommendationResult` : Structured output with explanations
- `BatchRecommender` : Efficient batch processing

```
engine = RecommendationEngine.load("checkpoint.pt")
result = engine.recommend(
    user_id=123,
    session_id="abc",
    item_history=[101, 205, 312],
    mood_text="I want something cozy",
    top_k=10
)
print(engine.explain_recommendation(result))
# Output:
# Based on your mood: "I want something cozy"
#   Detected mood: cozy
# Blending: 65% watch history + 35% current mood
# Top recommendations:
#   1. You've Got Mail (score: 0.892)
#   ...
```

8. `config.yaml` - Extended Configuration

Location: `Fusion/config.yaml`

New Settings Added:

```
# LLM Fusion settings
use_llm_fusion: True              # Enable LLM fusion
llm_dim: 768                      # Sentence transformer dimension
fusion_dropout: 0.1               # Dropout in fusion layers
vector_gate: True                 # Per-dimension gating

# Phase-specific learning rates
phase2_learning_rate: 0.0005
phase2_epochs: 50
phase3_learning_rate: 0.0001
phase3_epochs: 20
ewc_lambda: 0.4                   # EWC regularization strength
```

Summary

This integration plan provides:

1. **CE loss** as the recommended loss function for better gradient flow to all items
2. **Gated fusion** as an appropriate mechanism with suggestions for vector-level enhancement
3. **Gap analysis** of the current code identifying missing imports, data pipeline, and training loop modifications
4. **Refined training strategy** with contrastive pre-alignment and regularization
5. **Complete architecture** with database schemas for mood and profile storage
6. **Data flow diagrams** showing real-time inference, feedback loops, and training phases

The `Fusion/` folder contains all necessary code modifications to make LLM-Mamba integration fully functional.

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1. **CE loss** as the recommended loss function for better gradient flow to all items
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The next step is to create the `Fusion/` folder with all necessary code modifications.